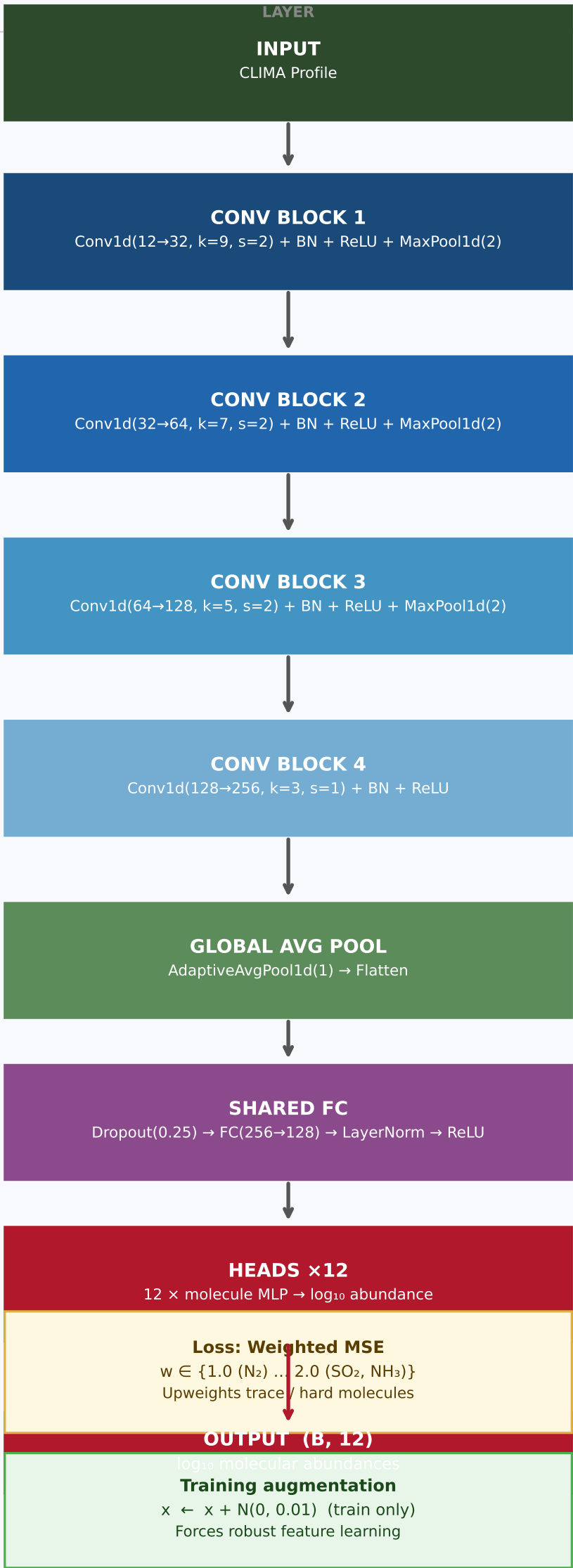


1D CNN — 1D Convolutional Network for Atmospheric Retrieval

Input: CLIMA profile (Batch × 12 channels × 101 altitude levels) → Output: 12 log₁₀ molecular abundances



OUTPUT SHAPE	DESCRIPTION
B, 12, 101	12 CLIMA channels × 101 altitude levels (Z-normalised per channel)
B, 32, 25	kernel=9 captures broad altitude patterns stride=2 + MaxPool halves resolution → 25 pts
B, 64, 6	kernel=7 resolution reduced to 6 pts channel depth doubled to 64
B, 128, 1	kernel=5 resolution collapsed to 1 pt channel depth 128
B, 256, 1	kernel=3, stride=1 no resolution change expands to 256 channels
B, 256	Squeeze altitude dimension → single 256-d vector per sample
B, 128	Shared representation for all 12 molecule heads (128-d)
B, 12	Each head: FC(128→hidden→1) output: log ₁₀ surface volume mixing ratio

